

Grey level Transformation

$S = 255 - x$

Pixel Values : $x = 50, 100, 150, 200$

Purpose:

Grey level is used to improve the appearance of an image by changing the intensity of pixel values. It helps in enhancing brightness, contrast

Negative Transformation: $S = 255 - x$

Original pixel	Calculation	Transformed pixel
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Histogram equalization

Project

Histogram equalization

Contrast of an image is spread
Pixel intensity value into a wide

Intensity	frequency
0	4
1	6
2	10
3	6

Total pixels = $N = 4 + 6 + 10 + 6 = 26$

Cumulative distribution function (CDF)

Intensity	frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

Equalized Intensities

$$S = \frac{CDF}{N} \times (L-1)$$

$$N=26$$

$$L=4$$

$$S = \frac{CDF}{26} \times 3$$

pdf.

Intensity	CDF	Calculation	Equalized Values
0	4	$(4/26) \times 3 = 0.46$	0
1	10	$(10/26) \times 3 = 1.15$	1
2	20	$(20/26) \times 3 = 2.31$	2
3	26	$(26/26) \times 3 = 3$	3

3. Gaussian Smoothing

a) Purpose:

Gaussian filtering removes noise while preserving edges better the average preserving it gives more weight of the central pixel.

(b) $\text{Kernel} = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ Image region $= \begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$

$$= (10 \times 1) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (40 \times 4) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (10 \times 1)$$

$$= 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10$$

$$= 360 \Rightarrow \frac{360}{16} = 22.5 \approx 23$$

New centre pixel = 23

4. Edge detection using Sobel operator

(a) Purpose

Edge detection identifies for sudden intensity change in an image. It helps detect object boundaries and shapes.

(b) Horizontal Sobel mask

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

new:

$$(-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -40$$

middle row:

0

bottom:

$$(1 \times 30) + (2 \times 30) + (1 \times 30) = 120$$

$$G = -40 + 120 = 80$$

Gradient = 80

Contrast Stretching

Purpose:

Contrast Stretching increases image contrast by expanding a narrow intensity range to the full grayscale range.

$$S = \frac{(x - x_{\min})}{(x_{\max} - x_{\min})} \times 255$$

$$x_{\min} = 50$$

$$x_{\max} = 150$$

$$\text{For } x = 50$$

$$S = \frac{50 - 50}{150 - 50} \times 255 = 0$$

$$T_{100} = 100$$

$$S = \frac{100 - 50}{100} \times 1255$$

$$= 0.5 \times 1255$$

$$= 1275 \times 0.5$$

$$T_{100} = 100$$

$$S = \frac{100 - 50}{100} \times 1255$$

$$= 0.5 \times 1255$$

$$= 255$$

Original (0)

Transformed (0)

50

0

100

125

150

255